

Part 2: Filtering Luminance (Y) Matrix Analysis

1. Mathematical Difference Between Filtering the Y Matrix vs. Filtering the R, G, and B Matrices Directly

When we filter the R (Red), G (Green), and B (Blue) channels of an image on their own, we average the pixel brightness in each color channel with neighboring pixels separately. If there's a sudden change in one color channel at an edge but not in the others, this independent filtering can mess up the original color balance (R:G:B) of the pixels.

In contrast, when we filter the Y (Luminance) channel, we're working with a single value that combines the color channels using the formula $Y = 0.299R + 0.587G + 0.114B$. This means the filter works on the overall brightness of the pixels rather than on each color separately. The direction of the color remains the same, but the strength or brightness is what gets adjusted.

2. Why Does Filtering the Y Channel Keep Colors Looking Natural?

Our eyes are more sensitive to brightness changes than to color changes. When we blur the R, G, and B channels separately, it can create unwanted bright edges called "color fringing." This happens because mixing the pixels creates new colors that don't fit with the original object.

When we only filter the Y channel, we smooth out the image's brightness and transitions without losing the original colors (hue and saturation). So, when we put the filtered Y channel back with the original colors, the image looks natural and perfect to us because the colors stay separate and do not blend incorrectly.